

Yuvraj Kashyap

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Software Engineer

EDUCATION

University of Texas at Dallas

Expected May 2027

B.S. in Computer Science, Minor in Finance, Certificate in Entrepreneurship & Innovation

GPA: 3.77/4.00 | Dean's List | NCAA Division II Tennis

Relevant Coursework: Advanced Algorithms, Data Structures & Algorithms, Database Systems, Systems Programming, Software Engineering, Programming Languages, Machine Learning, Artificial Intelligence, Computer Vision, Linear Algebra, Probability & Statistics

EXPERIENCE

Software Engineering Intern, Medceptor

Apr 2026 – Aug 2026

- Enabled CNA/LPN market expansion by shipping 14+ PRs for core question-bank and exam workflows used by paying, recurring users of Medceptor's medical education platform.
- Built workflows supporting 100+ generated questions, track/level filtering, import validation, scoring, and review using Next.js, TypeScript, Django, and Supabase/Postgres; carried features through testing and release QA.

Undergraduate Researcher, UAV / Smart City Systems

Oct 2025 – Present

University of Texas at Dallas

- Built a reproducible Python urban-geometry and UAV simulation framework from 1,553 OpenStreetMap building footprints across a 4 km × 4 km Dallas environment for path planning, visibility, coverage, and obstacle avoidance.
- Implemented 2.5D line-of-sight analysis, used greedy set cover to cover 224 of 240 sampled targets with six observers, and computed a 4.99 km eight-neighbor A* route around altitude-dependent obstacles.

Software Research Assistant, NOVA Autonomous Driving

Sep 2023 – Jun 2025

University of Texas at Dallas

- Built Python preprocessing pipelines for 3D sensor data, including voxel-grid point cloud downsampling that reduced scan size from 370,277 to 20,528 points while preserving geometric structure.
- Debugged power, sensor, and I/O reliability issues across autonomous vehicle subsystems, translating hardware constraints into software and systems-level fixes for perception reliability.

Vice President of Finance & Senior Consultant, Consult Your Community

Jan 2026 – Present

University of Texas at Dallas

- Built Glydr's web-based gaming hub prototype, translating client requirements into a working product and owning technical deliverables from scoping through final handoff.

PROJECTS

WorldState – GPU-Backed Real-Time Vision Runtime

Python, FastAPI, PyTorch, CUDA, OpenCV, WebSockets, PostgreSQL, React, Docker

- Built a local-first inference runtime coordinating vision models through asynchronous discovery, optical-flow propagation, shared-memory workers, and versioned REST/WebSocket contracts.
- Verified a 30-minute learned-model soak at 30.93 useful updates/s and 33.69 ms p95 with 2,588 MiB flat reserved VRAM, isolating storage latency from the frame-acknowledgement path.

Atlas – Fault-Tolerant Distributed Crawl, Index & Search Platform

Python, FastAPI, PostgreSQL, Redis/RQ, OpenSearch, Docker, AWS, Terraform, React, Vite

- Built PostgreSQL-authoritative scheduling, leased workers, heartbeats, bounded retries, dead-letter state, durable outbox indexing, deduplication, and service-health observability.
- Passed a deterministic 10,000-page fault benchmark covering worker loss, complete Redis data loss, retries, duplicate delivery, redirects, malformed content, and transient failures.

Aletheia – Hybrid Retrieval, Reranking & Search Evaluation Platform

Python, FastAPI, PyTorch, PostgreSQL, Redis/RQ, OpenSearch, Qdrant, Docker, Next.js

- Engineered a search and ranking platform comparing BM25, dense retrieval, hybrid Reciprocal Rank Fusion, and cross-encoder reranking with query tracing, experiment comparison, and reproducible evaluation workflows.
- Benchmarked 5,183 BEIR SciFact documents across 300 queries and 339 relevance judgments; hybrid RRF achieved 0.8429 Recall@10 with zero failed queries.

Beyond Chat – Artifact-Centric AI Workspace with Finance Agent

Live prototype

React, TypeScript, FastAPI, Supabase, OpenRouter, Exa, Stripe, LangChain, Vercel Sandbox, Pytest, Vitest

- Built a full-stack AI workspace for chat, writing, research, image, data, and finance workflows with workspace-scoped RLS, Stripe billing, artifact storage/search, streaming responses, and Dexter finance-agent execution through Vercel Sandbox.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C++, SQL

Backend & APIs: FastAPI, Django, Hono, Node.js, REST APIs, Server Actions, SQLAlchemy, Alembic, Pydantic, Zod

Data, Search & Infrastructure: PostgreSQL, Redis/RQ, OpenSearch, Qdrant, Supabase Auth/RLS, Docker, Docker Compose, Git, GitHub Actions, Vercel, Doppler, Linux/Unix, CI/CD

AI/ML Systems: BM25, vector search, embeddings, cross-encoder reranking, retrieval evaluation, OpenRouter, Exa, LangChain, PyTorch, NumPy, Pandas, OpenCV

Frontend & Integrations: React, Next.js, TanStack Query, Tailwind CSS, Vite, Framer Motion, Recharts, Stripe, Resend, QStash